

All powers of very general Jacobians of elementary abelian covers

Kyle Kistner

5 September 2026

Abstract

We give a proof of the rational Hodge conjecture for all powers of the very general Jacobian in a full branch family of elementary abelian covers of the projective line of fixed prime exponent. For odd primes, the proof combines an explicit classification of connected monodromy blocks with the balanced mixed-determinant fusion theorem of the companion manuscript. A rational algebraic correspondence from determinant-word Hodge structures spans the monodromy invariants. Semisimplicity then lifts Hodge classes to balanced sources, where fusion applies. This avoids the incorrect identification of the full monodromy commutant with Hodge endomorphisms and requires no formula for the central Hodge torus. The binary case uses independent symplectic factors and divisor classes. This is a written proof draft using precisely stated published inputs and the companion fusion proof; it is not a complete Lean formalization or an independently refereed result. Novelty is not asserted.

1 The theorem and its scope

Let p be prime, $G = (\mathbf{Z}/p)^m$ with $m \geq 1$, and let $a_1, \dots, a_r \in G \setminus \{0\}$ generate G and satisfy $\sum_i a_i = 0$. Assume $r \geq 3$. Let $B = M_{0,r}$ be the full space of ordered distinct branch points modulo projective transformations. The specified local monodromies define a family of connected G -covers of $\mathbf{P}^1$, interpreted on a finite cover when necessary to choose a family and auxiliary graphs. Write X_b for a fiber and $A_b = J(X_b)$. The full branch family is essential: an arbitrary subfamily is not substituted for B .

Theorem 1.1. *There is a countable union $Z \subset B$ of proper closed algebraic subsets such that, for every $b \in B(\mathbf{C}) \setminus Z$, every $q \geq 1$, and every $d \geq 0$, the cycle class map*

$$\mathrm{CH}^d(A_b^q)_{\mathbf{Q}} \longrightarrow H^{2d}(A_b^q, \mathbf{Q}) \cap H^{d,d}(A_b^q)$$

is surjective. If $p = 2$, these Hodge classes are generated by divisor classes.

For $r = 2$ the only connected data generate a cyclic group and the curve has genus zero, so the conclusion is immediate. For $r = 3$ the parameter space is a point; the odd-prime proof below includes its constant rank-one factors. The theorem concerns rational classes over $\mathbf{C}$. It makes no claim about integral classes, special fibers with additional Hodge classes, composite exponent, nonabelian groups, or explicit equations for all cycles.

The companion manuscript [2], Theorem 1.1, proves algebraicity of balanced mixed determinants for *arbitrary* finite lists of cyclic p -covers of $\mathbf{P}^1$, with repetitions and specified generators. That result uses neither this theorem nor a general monodromy/generator hypothesis. Its final conditional all-powers application is replaced here by an explicit generator proof. The original frozen Phase I and Phase II files are not premises of this argument.

2 Character factors and a mixed embedding

Suppose first that p is odd. Put $K = \mathbf{Q}(\zeta_p)$ and $\Gamma = \mathbf{F}_p^\times = \text{Gal}(K/\mathbf{Q})$. The evaluation code is

$$C = \{(c(a_1), \dots, c(a_r)) : c \in \text{Hom}(G, \mathbf{F}_p)\} \subset \mathbf{F}_p^r.$$

Generation of G makes evaluation injective. We use the same symbol c for a character and its word. Write $S(c) = \{i : c_i \neq 0\}$, $s(c) = |S(c)|$, and $h(c) = s(c) - 2$. Over K or $\mathbf{C}$, let V_c be the $\zeta_p^{c(g)}$ character part of $V = H^1(A_b, \mathbf{Q})$. A uniform reversal of this convention has no effect on the conclusions.

The quotient $Y_c = X_b / \ker c$ is a cyclic p -cover with active residues $(c_i)_{i \in S(c)}$. Its positive-genus rational factor $U_c = H^1(JY_c, \mathbf{Q})$ has K -rank $h(c)$, and

$$U_c \otimes_{\mathbf{Q}} \mathbf{C} = \bigoplus_{t \in \Gamma} V_{tc}.$$

One representative from each nonzero character line gives the rational isogeny decomposition of A_b . Pullback and norm realize the associated rational cohomological inclusions and projections algebraically. Support two has genus zero; support one is impossible by the sum-zero condition. We use only supports at least three below.

Lemma 2.1 (A mixed embedding always exists). *For nonzero residues $A_1, \dots, A_s \in \mathbf{F}_p$ with sum zero and $s \geq 4$, there is $t \in \Gamma$ such that*

$$2 \leq Q(t) := \frac{1}{p} \sum_i [tA_i]_p \leq s - 2.$$

Consequently both Hodge multiplicities of the corresponding character space are positive at some embedding.

Proof. The integers $Q(t)$ lie between 1 and $s - 1$, and $Q(p - t) = s - Q(t)$. If $Q(1)$ is already interior there is nothing to prove. Otherwise negate the word to arrange $Q(1) = 1$, and use positive representatives A_i with $\sum A_i = p$. For $1 \leq t \leq p - 2$,

$$Q(t + 1) - Q(t) = 1 - \#\{i : [tA_i]_p + A_i \geq p\} \leq 1.$$

No endpoint residue is zero in this range. Since $Q(p - 1) = s - 1 \geq 3$, the first crossing of 2 equals 2. The two Hodge dimensions are $Q(t) - 1$ and $s - 1 - Q(t)$, in one order, by the cyclic eigendifferential formula [1, Theorem 2.1]; see also [2, Section 6]. $\square$

Proposition 2.2. *Let M be the connected algebraic monodromy group on V . Every complex projection of M on V_c with $s(c) \geq 4$ is $\text{SL}_{h(c)}$. This includes the definite conjugate embeddings. The only nonzero character factors with trivial connected monodromy have support three and rank one.*

Proof. Choose an active point as infinity and set $n = s - 1$. At the embedding of Lemma 2.1, the sum of finite fractional weights is $Q(t) - [tA_\infty]_p/p$, strictly between 1 and $n - 1$. All finite weights are active, $n \geq 3$, and the gcd condition is automatic at a prime. Thus Definition 1.1(a) and Theorem 5.1 of [1] give full special unitary projection, whose complexification is SL_{s-2} , on the affine configuration space $\text{Conf}_{s-1}(\mathbf{C})$. Normalizing two finite points to 0, 1 identifies that space with $\text{Aff}_1 \times M_{0,s}$. The change $x = \alpha z + \beta$ rescales the Kummer coordinate by $\alpha^{(\sum_{i \neq \infty} [A_i]_p)/p}$; thus

the affine-group fiber has only finite scalar monodromy on each character. Passing to $M_{0,s}$ preserves the connected closure. Finally $M_{0,r} \rightarrow M_{0,s}$ has connected configuration-space fibers and surjects on fundamental groups. This transfers the conclusion to the full branch family without asserting a surjection from $\pi_1(M_{0,s})$ to an affine pure braid group.

In a K -basis of the eigenspace in $V \otimes_{\mathbf{Q}} K$, monodromy matrices have entries in K . The Zariski closure of a set of K -points commutes with extension from K to $\mathbf{C}$: expand the coefficients of a vanishing complex polynomial in a finite K -linearly independent list and evaluate at the K -points. Each resulting K -coefficient polynomial vanishes separately. Applying $\zeta_p \mapsto \zeta_p^u$ therefore transports the closure to that on V_{uc} . Full SL_h at one embedding gives full SL_h at every embedding. Pair-braid determinants and the deck scalar ambiguities have finite order [1, Corollary 2.7], so there is no connected scalar factor in this argument.

With three active points, normalization to $0, 1, \infty$ makes the cyclic family isotrivial after a finite root choice. Its monodromy is finite. The positive-rank cases have now all been accounted for. A definite complex eigenspace alone has no rational integral lattice; compactness at that one embedding does not contradict its full complex Zariski closure. $\square$

3 Exact monodromy blocks and actual oriented graphs

For four labelled positions S , let $\mathcal{V}_4(S)$ be the Klein four group of double transpositions and the identity. Orbit notation below always means intersection with the evaluation code C ; C need not be preserved by all these permutations.

Theorem 3.1. *On the indices $c \in C$ with $s(c) \geq 4$, connected monodromy has the following complex simple blocks:*

1. *Different active supports belong to independent factors.*
2. *On a fixed support of size at least five, a block is $\{c, -c\}$, with standard and dual representations of $\mathrm{SL}_{h(c)}$.*
3. *On a support of size four, a block is $(\mathcal{V}_4(S)c \cup -\mathcal{V}_4(S)c) \cap C$. Its oriented component $O_c = (\mathcal{V}_4(S)c) \cap C$ consists of actual algebraically isomorphic character spaces; $-O_c$ is its polarized dual component. The two components may coincide.*

The connected linear image is the product of these SL_h factors with the indicated occurrences. The entire description is Galois-stable.

3.1 The same projective group before finite base change

Projective character monodromy is defined on the same group $\Pi = \pi_1(M_{0,r})$ for every c : two lifts of a marked-sphere mapping class differ by a deck transformation, hence by a scalar on V_c . This gives a product of projective representations. It does not claim one common scalar on their direct sum. Root choices or level covers restrict to finite-index subgroups and leave connected closures unchanged.

For a centerless simple algebraic group H and an automorphism θ ,

$$N_{H \times H}(\mathrm{Graph}(\theta)) = \mathrm{Graph}(\theta). \quad (1)$$

Indeed a normalizing pair (a, b) makes $\theta(a)^{-1}b$ centralize H . The connected component is normal in the full closure. Consequently, when a pair of connected projective projections is a graph, *every*

original braid image belongs to that graph. We may compare finite-order pair braids before finite base changes raise their eigenvalues to powers.

The connected solvable radical of $M_{\mathbf{C}}$ projects to a connected solvable normal subgroup of each full SL_h , hence trivially. It also acts trivially on the finite-monodromy summands. Faithfulness makes the radical trivial. Semisimple Goursat theory now partitions the projective projections into graph blocks. At the Lie-algebra level, each simple target receives exactly one simple ideal: two commuting nonzero ideals cannot both surject onto it. In particular there are no extra relations invisible in pairwise projections.

3.2 Nonidentity twists, including zero pair sums

Let i, j be active for c , with $s(c) \geq 4$, and choose another active label as infinity. Put $n = s(c) - 1$; the finite residue sum is nonzero and $\dim V_c = n - 1 \geq 2$. Number the coordinate description so the chosen pair is $(1, 2)$; this relabels a description, not an action by a pure braid. Theorem 2.2 of [1] gives n spanning vectors $g_1, \dots, g_n$ with their only relation $\sum_i g_i = 0$. Thus $g_1 \neq 0$. Theorem 2.5 there gives

$$\rho_c(A_{12}) - I = \kappa g_1 \otimes \langle -, g_1 \rangle, \quad \kappa \neq 0,$$

where the Hermitian form is nondegenerate and the coefficient is nonzero because both local residues are active. Hence the difference has rank one and is nonzero. It is not a scalar difference in dimension at least two. Every active pair twist is therefore projectively nontrivial.

For the chosen positive-eigenvalue character convention, put $q = \zeta_p$ in the source pair formula. If $c_i + c_j \neq 0$, its spectrum is

$$(\zeta_p^{c_i+c_j}, 1, \dots, 1). \quad (2)$$

If the sum is zero, the same rank-one argument proves it is a *nonidentity* unipotent. This last conclusion cannot be inferred from the word “unipotent” alone. For example, at $p = 3$ the zero-sum word $(1, 2, 2, 2, 2)$ has zero sum on every pair containing its first position.

If $S(c) \neq S(e)$ and their ranks agree, choose $i \in S(c) \setminus S(e)$ and $j \in S(c) \setminus \{i\}$. The same pair twist is nontrivial on V_c , but becomes trivial projectively on V_e after forgetting i . This contradicts (1). Unequal ranks already give nonisomorphic simple groups. This proves support separation.

3.3 Reconstruction on a fixed support

For $h \geq 3$, every complex automorphism of PGL_h is inner or inner followed by inverse transpose. A graph therefore has a single fixed sign $\epsilon \in \{1, -1\}$. The multiplicities $1, h - 1$ in (2) distinguish the exceptional eigenvalue projectively. Comparison of all finite pair braids, including the unipotent case, gives

$$e_i + e_j = \epsilon(c_i + c_j).$$

For three distinct finite indices,

$$2e_i = (e_i + e_j) + (e_i + e_k) - (e_j + e_k) = 2\epsilon c_i.$$

Since p is odd, all finite coordinates agree up to ϵ , and the sum-zero relation gives the infinity coordinate. Thus $e = \epsilon c$. The converse follows from identity and polarization.

For $h = 2$, write the four active coordinates as c_1, c_2, c_3, c_4 and put $x = c_1 + c_2$, $y = c_1 + c_3$, $z = c_1 + c_4$. The inverse is

$$c = \frac{1}{2}(x + y + z, x - y - z, -x + y - z, -x - y + z). \quad (3)$$

All automorphisms of PGL_2 are inner. The unordered eigenvalue ratios of the three finite pair braids give independent signs on $x, y, -z$. Thus $(x_e, y_e, z_e) = (\epsilon_x x, \epsilon_y y, \epsilon_z z)$. The four even sign changes are exactly the effects of id , $(12)(34)$, $(13)(24)$, $(14)(23)$; the other four are their negatives. Equation (3) proves $e = \epsilon Pc$. Zero pair sums only enlarge stabilizers and give no additional solutions.

For the geometric converse, normalize the four branch points to $0, 1, t, \infty$. The corresponding double transpositions are

$$T_{12|34}(x) = \frac{t(x-1)}{x-t}, \quad T_{13|24}(x) = \frac{x-t}{x-1}, \quad T_{14|23}(x) = \frac{t}{x}.$$

Write the cyclic equations as $y^p = f_c(x)$ and $z^p = f_e(x)$. For $e = Pc$, the divisor of $f_e(T_P(x))/f_c(x)$ is divisible by p . A degree-zero divisor on $\mathbf{P}^1$ is principal, so

$$f_e(T_P(x))/f_c(x) = a(t)h(x)^p.$$

Here $a(t)$ is a nonzero constant times powers of t and $t-1$; this also follows directly from the displayed transformations. After adjoining $a(t)^{1/p}$, a finite étale cover of $M_{0,4}$, the map

$$(x, y) \longmapsto (T_P(x), a(t)^{1/p}h(x)y)$$

is deck-equivariant and extends to an isomorphism of the smooth projective curves. Its graph, or inverse graph as appropriate for cohomological variance, identifies V_{bc} with V_{be} for *every* $b \in \Gamma$. For the full family pull it back along the active-point forgetting map. Pullback and norm for the cyclic quotients transfer these graphs to the rational factors of A_b .

The negative component is obtained using the polarized identification $V_{-c} \simeq V_c^*(-1)$, together with these oriented graphs. At the level of SL_2 representations the dual is isomorphic to the standard, which gives the claimed adjoint graph. It does not assert a degree-zero Hodge isomorphism between an arbitrary V_c and V_{-c} . If $O_c = -O_c$, a double transposition sends c to $-c$; its two pairs are opposite residues, and every embedding has signature $(1, 1)$. The oriented graph to the negative character is then actual, as asserted.

These arguments give all necessary and sufficient pairwise graph relations. Goursat supplies the product. Finite discrepancies between linear lifts vanish on the connected group; its Lie algebra and the standard/dual representations lift to the product of the simply connected groups SL_h . Each factor has a faithful standard occurrence, so its connected linear image is exactly $M_{\mathbf{C}}$. Multiplying all residues by a unit preserves every displayed relation. This completes Theorem 3.1.

Remark 3.2 (A false endomorphism shortcut). At $p = 5$, the word $(1, 1, 1, 2)$ has signatures $(2, 0)$, $(1, 1)$, $(1, 1)$, $(0, 2)$, up to convention. Its opposite character is isomorphic as an SL_2 module, but at a definite embedding has the opposite pure Hodge type. The nonzero monodromy Hom there is not a Hodge Hom . We do not identify $\mathrm{End}_M(V)$ with the Hodge endomorphism algebra in general.

4 Determinant words and algebraic rational projectors

For each positive-genus character c , choose a deck generator on Y_c whose induced action has eigenvalue ζ_p^b on V_{bc} . Equivalently choose $g_c \in G$ with $c(g_c) = 1$. Different scalar multiples of c may use the same curve with different generators; they remain distinct word entries. A nonempty effective

word $w = (c_1, \dots, c_N)$ permits repetitions. Put $H = \sum_j h(c_j)$. Its rational determinant source is

$$W_w \otimes_{\mathbf{Q}} \mathbf{C} = \bigoplus_{b \in \Gamma} \bigotimes_{j=1}^N \bigwedge^{h(c_j)} V_{bc_j} \subset H^H \left(\prod_j JY_{c_j}, \mathbf{C} \right). \quad (4)$$

It has rational dimension $p - 1$ and rank one over K . This is the matching-embedding tensor product over K , not the full tensor product of rational determinant spaces.

We recall explicitly why its realization and the operations used below are rational algebraic. On the full Jacobian the cyclic-line projector is

$$\Pi_c = \frac{1}{|\ker c|} \sum_{g \in \ker c} g^* - \frac{1}{|G|} \sum_{g \in G} g^*.$$

Its image is U_c ; the subtracted invariant H^1 is zero here. On this image put $T_c = g_c^*$. For raw H^1 slots with intended character labels $c_1, \dots, c_n$, first apply their Π_{c_i} and then

$$E(c_1, \dots, c_n) = \prod_{i=2}^n \left(\frac{1}{p} \sum_{a=0}^{p-1} T_{c_1,1}^a T_{c_i,i}^{-a} \right). \quad (5)$$

A root-of-unity sum shows that its image is exactly $\bigoplus_b \bigotimes_i V_{bc_i}$. Each term is a rational algebraic deck correspondence. On identical slots the matching projector commutes with ordinary permutations, since equality of all embedding labels is permutation-invariant.

The degree projectors on an abelian variety are cohomologically algebraic: for any integer $m \geq 2$, interpolate the distinct eigenvalues m^i of $[m]^*$ to obtain

$$\pi_k = \prod_{\substack{0 \leq i \leq 2 \dim D \\ i \neq k}} \frac{[m]^* - m^i}{m^k - m^i}.$$

For addition $\mu : D^n \rightarrow D$, diagonal $\delta : D \rightarrow D^n$, and $P = \pi_1^{\otimes n}$, set $\iota_n = P\mu^*$ and $q_n = \delta^*P$. On the relevant degree summands,

$$q_n \iota_n = n! \text{id}, \quad \iota_n q_n = n! \text{Alt}_n.$$

Here Alt_n is the ordinary antisymmetrizer. Ordinary permutations are realized by geometric factor permutations with their degree-one Koszul sign corrected. Thus insertion of alternating volume tensors, and passage back to exterior cohomology, are algebraic correspondences with explicit nonzero rational normalizations.

Apply (5) to all raw word slots, alternate within each volume block, and return using $q_{h(c_j)}$ and the factorials. This constructs an algebraic rational projector with image (4). On the matched image, action by ζ_p on *one raw slot* acts by ζ_p^b on its b -line and commutes with the alternating restrictions. It gives a faithful K -action by algebraic Hodge endomorphisms on W_w . Simultaneous deck action on all $h(c_j)$ slots would instead give $\zeta_p^{bh(c_j)}$, possibly 1 when $p \mid h(c_j)$; that incorrect action is not used. These constructions are also detailed in [2, Section 2].

Proposition 4.1 (The source Hodge criterion). *Suppose $H + 2t = 2d$, where $t \geq 0$. Then the rational Hodge-class space of $W_w \otimes \mathbf{Q}(-t)$ is either zero or the whole source. It is the whole source precisely when*

$$2 \sum_{j=1}^N \sum_{i \in S(c_j)} [bc_{j,i}]_p = p \sum_{j=1}^N s(c_j) \quad (b \in \Gamma). \quad (6)$$

In that case every source Hodge class is algebraic, in its specified cohomological realization with the Tate factor.

Proof. The rational Hodge-class subspace is stable under the K -action by Hodge endomorphisms. Since W_w is a K -line, a nonzero such subspace is the whole line. Equivalently, a nonzero rational vector has nonzero components at all embeddings of K .

Let $d_c(b) = \dim V_{bc}^{1,0}$. The b -line in (4) has type $(\sum_j d_{c_j}(b), H - \sum_j d_{c_j}(b))$. After twisting by $\mathbf{Q}(-t)$ it is of type (d, d) exactly when $\sum_j d_{c_j}(b) = H/2$. The cyclic dimension formula gives (6); the opposite convention merely replaces b by $-b$. Here H is even since $H + 2t = 2d$. The standalone balanced fusion theorem [2, Theorem 1.1] applies to this exact word, including repeated curves, rank-one factors and chosen generators. It makes all of W_w algebraic. Tate factors are represented by products of point or divisor classes. No genericity is required for this last application. $\square$

5 Rational diagrams span every monodromy invariant

We prove the generator step rather than treating it as a hypothesis about the central torus. All arguments are in a fixed finite tensor degree.

Lemma 5.1 (Mixed tensor invariants). *Let U have dimension $h \geq 2$ over a characteristic-zero field. Invariants for $\mathrm{SL}(U)$ in tensor products of standard and dual copies are spanned by diagrams made from coevaluations, alternating h -slot standard volumes, and alternating h -slot dual volumes.*

Proof. We may extend scalars to an algebraic closure, since invariants and the claimed linear span commute with extension. For a standard and b dual slots, the center μ_h forces $a - b$ to be divisible by h . If $a = b + mh$ with $m \geq 0$, the invariant space, after choosing a nonzero volume in $\det(U)^m$, identifies with

$$\mathrm{Hom}_{\mathrm{SL}(U)}(U^{\otimes b}, U^{\otimes a}) \simeq \mathrm{Hom}_{\mathrm{GL}(U)}(U^{\otimes b} \otimes \det(U)^m, U^{\otimes a}).$$

The scalar degrees on the latter source and target agree, so SL equivariance gives GL equivariance. Alternating m disjoint blocks of h slots embeds that source as a GL -equivariant direct summand of $U^{\otimes a}$. Extend a map using the alternating projector. Schur–Weyl duality [6, Theorem 4.57(i), Proposition 4.58] says every GL -equivariant endomorphism of this tensor power is a linear combination of slot permutations. This produces m volumes and b coevaluations. If $a < b$, interchange U and U^* . The argument is valid also when $h < a$; permutation operators need span, not be linearly independent. $\square$

Fix $q \geq 1$ and $n = 2d$. Apply the lemma to the product of blocks in Theorem 3.1, with occurrences from $H^1(A_b^q)$ labelled separately. Within an oriented component, use its actual graph maps to identify the standard copies. The negative component is the polarized dual $U^*(-1)$, so a standard–dual coevaluation has source $\mathbf{Q}(-1)$ and is a divisor tensor. A same-orientation volume has source $\det U$; it is not assumed Hodge. Dual volumes use the actual negative-character determinant, of weight h , including its Tate bookkeeping.

For a self-dual rank-two component $O_c = -O_c$, one may identify every copy as standard by the actual graphs and use two-slot volumes. Keeping them as determinant sources is harmless. Support-three factors have rank one and trivial connected monodromy; each of their slots is itself a one-slot determinant source. Thus no constant CM part is omitted. Every diagram has a nonempty word w of total volume degree H and t pairing blocks with $H + 2t = n$, or has an empty word and consists entirely of pairings. All multiplicities are nonnegative.

Lemma 5.2 (Algebraic rational diagram realization). *For each such complex diagram and its full Galois orbit there is a rational algebraic Hodge map into $H^n(A_b^q, \mathbf{Q})$ from $W_w \otimes \mathbf{Q}(-t)$, or from a finite sum of $\mathbf{Q}(-t)$ for an empty word, whose scalar extension contains every diagram in that orbit in its image. Collectively these maps have image exactly $H^n(A_b^q, \mathbf{Q})^M$.*

Proof. First work in raw tensor slots, with intended labels $c_1, \dots, c_n$. For each volume choose its representative character a_j and its cyclic quotient Y_{a_j} . The alternating inclusion from its determinant source, followed by the actual oriented graphs and quotient pullbacks, maps its b -line to the prescribed slots $V_{b c_i}$. The maps work simultaneously for every b ; no individual complex eigenprojector is called rational.

For each pairing use the rational polarization coevaluation on the corresponding cyclic factor. Its extension is the sum of the nonzero coevaluations in $V_{ba} \otimes V_{-ba}$, for $b \in \Gamma$. A Poincaré divisor, algebraic degree projections, and rational inverse isogeny for the polarization realize this tensor algebraically. Apply the actual oriented graphs to the two slots. This produces precisely the conjugates of the desired standard–dual pairing with nonzero compatible normalizations.

For a nonempty word, append the t rational pairings to W_w , reorder the slots, and apply the *global* projector $E(c_1, \dots, c_n)$ from (5). Before this last operation the pairing blocks may have independent embedding labels. The projector forces all their labels to equal the word label. Consequently we obtain a rational algebraic Hodge map

$$F_D : W_w \otimes \mathbf{Q}(-t) \longrightarrow H^1(A_b^q, \mathbf{Q})^{\otimes n}$$

whose restriction to the b -line is the b -conjugate diagram. It is nonzero on that tensor diagram before the exterior projection. Scalar extension separates the individual embedding lines, so its image contains each of them.

For an empty word with $n > 0$, apply the same matching projector to the external product of the t rational pairings. This gives an algebraic tensor equal to the sum of the conjugate diagrams. Apply $T_{c_1,1}^a$ for $0 \leq a \leq p-2$. These are $p-1$ algebraic maps from $\mathbf{Q}(-t)$. Their matrix on the b -diagrams is $(\zeta_p^{ab})_{b \neq 0, 0 \leq a \leq p-2}$, which is invertible: a polynomial of degree at most $p-2$ vanishing at all $p-1$ primitive p th roots is zero. Thus the full orbit, not just its trace, is spanned. For $n = 0$, use the unit $\mathbf{Q} \rightarrow \mathbf{Q}$.

All slot operations are algebraic by Section 4. Composing with the exterior/cup-product projection gives the required target maps. Diagrams that vanish or become dependent there do not matter: the antisymmetrizer supplies an equivariant section of the exterior quotient, so invariant tensor diagrams still span its invariant quotient.

There are only finitely many labels, partitions and slot permutations in degree n . Include a finite spanning collection and its complete Galois orbits. Each source is fixed by connected monodromy (determinants of SL_h , constant rank-one factors, and pairings), and the chosen graph maps commute with monodromy after a common finite base change. The images therefore lie in the invariants. Conversely the tensor lemma spans them after extension to $\mathbf{C}$. Invariants and images commute with scalar extension in these finite-dimensional rational representations, giving exact equality over $\mathbf{Q}$. $\square$

Remark 5.3. An abstract subdirect-product classification without the actual orientations would not suffice. Take a common SL_2 standard module U with four distinct central twists χ_i satisfying $\sum_i \chi_i = 0$ but no opposite pair. The product of alternating pairings in four different copies can be invariant, while individual determinants and actual Hodge-Hom maps have even signed multiplicity in each copy and cannot produce it. The geometric graph construction in Section 3 excludes these unrelated twists within an oriented component. We never promote an arbitrary projective monodromy graph to a Hodge correspondence.

6 Lifting Hodge classes and completing the odd-prime proof

The preceding construction yields a finite rational algebraic Hodge map

$$F : \mathcal{S} = \left(\bigoplus_{\mathcal{D}} W_{w_{\mathcal{D}}} \otimes \mathbf{Q}(-t_{\mathcal{D}}) \right) \oplus (\text{Tate summands}) \longrightarrow H^{2d}(A_b^q, \mathbf{Q}), \quad (7)$$

with image $I = H^{2d}(A_b^q, \mathbf{Q})^M$. Every summand has weight $2d$. The image is a rational Hodge substructure because F is a Hodge map; this conclusion does not require a separate assertion about the Hodge-group center.

Lemma 6.1 (Hodge lifting). *Every rational Hodge class in I has a rational Hodge preimage in $\mathcal{S}$.*

Proof. Polarizable pure rational Hodge structures are semisimple. Concretely, a polarization makes the orthogonal complement of $\ker F$ a Hodge complement on which F is an isomorphism onto I . Its inverse lifts the class. This splitting is only a Hodge splitting. No algebraicity of the splitting is assumed or used. $\square$

At a very general point, all rational Hodge classes on all powers lie in connected monodromy invariants. Here is the precise genericity input. The standard generic monodromy inclusion is [3, Section 4, Lemma 4]. For each power and degree, twist $R^{2d}f_*\mathbf{Z}$ by (d) to obtain the weight-zero variation (modulo torsion) with its polarization. Algebraicity of the exceptional Hodge loci follows from [4, Theorem 1.1 and Corollary 1.2]: bounded integral Hodge classes form an algebraic space finite over the base. Remove the proper images of its components, for every bound and each cohomological degree of each power. There are countably many such proper closed algebraic subsets. At the remaining points any Hodge class is generic and its monodromy orbit is finite; equivalently, connected monodromy fixes it. One can see the finiteness directly from the positive polarization on a locally constant type- $(0, 0)$ integral Hodge subsystem. Finite auxiliary base covers preserve this conclusion and the meaning of very general after taking images of the exceptional sets.

Let $\alpha \in H^{2d}(A_b^q, \mathbf{Q})$ be a Hodge class at such a point. It belongs to I . Lift it by Lemma 6.1 to a Hodge vector in (7). Its components are Hodge vectors separately in the displayed rational direct summands. By Proposition 4.1, a nonzero component in a nonempty word forces that entire word to satisfy all-row balance, and fusion supplies algebraic representatives. The Tate components are algebraic too. All *forward* maps in F are algebraic correspondences, external products with algebraic classes, and pullbacks or pushforwards. They therefore carry those representatives to algebraic cycles with class α . This proves Theorem 1.1 for odd p .

Relations and cancellations among complex diagram monomials cause no gap: we lift through a rational Hodge surjection, rather than assuming an arbitrary complex expansion of α is termwise balanced. The proof also uses all effective words needed in each fixed degree, not an integer basis of a lattice that might require negative multiplicities.

7 The binary case

For $p = 2$, a nonzero character is determined by its support, whose size s is even. Its cyclic quotient is hyperelliptic of genus $g = (s - 2)/2$. Supports of size two vanish; the remaining ones have $s \geq 4$. Their rational H^1 factors have connected monodromy Sp_{2g} . For precise application of A'Campo [5, Theorem 1, p. 319], put all $s = 2g + 2$ active branch points in the affine line. His parameter $\mu = s - 1 = 2g + 1$ is odd and at least three, including $g = 1$. The symplectic module is the homology of the affine curve modulo its intersection radical, hence the homology

of the compact curve. The full braid image contains the principal level-two subgroup. Passing to ordered configurations is finite index, so the connected Zariski closure remains Sp_{2g} . Passing from affine ordered configurations to $M_{0,s}$ has only finite deck-scalar fiber monodromy: lifts of the same marked-sphere map differ by a deck transformation. Thus normalization also preserves the connected closure. Pullback to $M_{0,r}$ uses the connected fibers of the active-point forgetting map. No claim about equality of the discrete image after arbitrary finite base change is needed.

Distinct supports give independent factors. Indeed the pair twist for two active points has a nonzero Picard–Lefschetz transvection (squared for ordered points). Its vanishing cycle is nonseparating: the double cover of a disk containing those two branch points is an annulus; the double cover of the complementary disk is connected because it contains the remaining $s - 2 \geq 2$ branch points. Cutting the annular core therefore leaves the whole covering surface connected. Thus its homology class is nonzero. If one of the two points is inactive for a different support, that twist becomes projectively trivial on the latter quotient by forgetting it. The normalizer argument (1), now for the adjoint symplectic group, and semisimple Goursat theory rule out a diagonal graph. Different genera already give nonisomorphic simple groups.

The symplectic tensor invariant theorem [7, Section 7.3, Theorem 7.4 and Corollary 7.5] says invariants in all tensor powers of standard modules are spanned by products of pairings. For this product of independent symplectic factors, each pairing is algebraic: use the quotient polarization and its Poincaré divisor, pulled back to the chosen copies of A_b . The same algebraic exterior projection gives all invariants in $H^*(A_b^g)$. Very-general Hodge classes are invariant by the argument in Section 6; hence they are generated by divisor classes. This completes the binary case.

8 Dependencies and verification boundary

The new universal arguments here are the mixed-embedding carry lemma, monodromy support and labelled-pair reconstruction, oriented graph construction, rational diagram realization, and Hodge lifting to balanced sources. The following inputs have distinct roles:

- Menet–Nguyen, Definition 1.1(a), Theorems 2.1, 2.2, 2.5, 5.1, and Corollary 2.7: cyclic dimensions, pair operators, and full factor projections. All hypotheses used are matched in Sections 2–3.
- The companion standalone fusion theorem: algebraicity only after a nonempty effective determinant word has all-row balance. Its inputs are the prime balanced-tuple theorem, Schoen’s simple-tuple cycles with the 1998 addendum scope, algebraic admissible-cover smoothing, compact-type Jacobians and cycle specialization. Their locators and hypotheses are recorded in that manuscript and its source audit.
- Classical automorphisms of adjoint linear groups, semisimple Goursat, Schur–Weyl duality and symplectic tensor invariants. The particular mixed SL tensor statement is proved from Schur–Weyl in Lemma 5.1; no finite computation replaces it.
- Generic monodromy and algebraic Hodge loci, with the polarization complement argument for Hodge semisimplicity.
- A’Campo and Picard–Lefschetz for the binary case.

The repository’s exact finite arithmetic checks exercise the carry formula, rank-two sign cube, higher-rank reconstruction, and several false shortcuts. They are regression evidence, not proofs of universal quantifiers or of cycle algebraicity. Its existing Lean results cover only designated

arithmetic and abstract interfaces; this entire geometric argument is not formalized. In particular, resolving this written all-powers argument does not repair the legacy complete Hodge-group or central-torus formulas. Independent specialist review of this manuscript together with the fusion companion remains necessary before treating the result as publication-ready.

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